Coverage Is Not Predictive Fidelity: A Split-Landscape Analysis of Data Condensation for Gradient-Boosted Trees

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***Abstract*—**
Data condensation replaces a large training set with a small surrogate, but tabular condensers are usually judged only after retraining a model on each candidate. For histogram gradient-boosted decision trees (GBDTs), we ask whether a candidate preserves the split decisions the full-data booster would make. We introduce Split-Landscape Discrepancy (SLD), a no-candidate-retraining diagnostic that compares full and condensed split-gain landscapes under fixed preprocessing and prediction state.

Across a broad audit of tabular condensers, method-level split-regret is highly correlated with downstream area under the receiver operating characteristic curve (AUROC) in the small-to-mid-size regime, while a separate core grid supports the headline performance comparisons. SLD is therefore best used as a rejection filter and method-family diagnostic, not as a selector for the single best method on a new dataset. It also exposes why coverage can fail: a greedy high-gain threshold-coverage selector can perform poorly because unweighted coverage differs from gain-weighted structural fidelity. At very large scale, random sampling becomes hard to beat and root-regret loses resolution as reasonable surrogates clear dominant split margins. We release the benchmark, scripts, processed tables, and diagnostic.

***Index Terms*—data condensation; gradient-boosted decision trees; coreset selection; tabular data; no-candidate-retraining diagnostics**

I. Introduction

ML engineers and data scientists who train gradient-boosted models on tabular data face a recurring operational problem: every candidate condensed dataset is usually judged by retraining a model on it. That is acceptable for a few surrogates, but it becomes costly when condensation is used inside repeated model development, hyperparameter search, data versioning, or governance workflows. A small surrogate can make these workflows cheaper and easier to share, although condensation is not itself a privacy or compliance guarantee. The applied question is therefore not only *which* surrogate gives the best final AUROC, but which candidates are worth retraining on at all.

Existing condensation and coreset methods provide many ways to build such surrogates [1]–[4], including recent methods designed for tabular data [5], [6]. Their common evaluation protocol, however, remains largely black-box: train a fresh model on each condensed set, measure downstream predictive performance, and compare. This is the right final validation step, but it is a poor diagnostic. It does not say whether a bad surrogate failed because it missed important split thresholds, distorted leaf statistics, or simply lacked enough budget. For practitioners, that distinction matters: a useful diagnostic should reject clearly bad surrogates early and explain which failure mode is being observed.

Histogram gradient-boosted decision trees (GBDTs) offer a concrete handle for such a diagnostic. At each node, a GBDT chooses the feature and threshold that maximise a regularised split-gain criterion [7]–[9]. Given a fixed prediction state, the candidate split gains are determined by binned sums of first- and second-order loss statistics. The resulting table of gains is the model's *split-gain landscape*: it records which structural decisions the booster is likely to make before leaf values are estimated. If a condensed set preserves this landscape, it has a plausible route to preserving the learned tree structure; if it distorts the landscape, downstream retraining is already suspect.

We introduce *Split-Landscape Discrepancy* (SLD), a no-candidate-retraining diagnostic that compares the split-gain landscape of the full data with that of a candidate surrogate under the same preprocessing and fixed prediction state. SLD is not a claim that no model state is ever used; rather, it replaces retraining on every candidate with a histogram-pass screening measurement. In an applied pipeline, its role is to flag high-risk surrogates and compare method families before spending retraining compute on the final candidates.

Using SLD as a measurement instrument, we evaluate 17 condensation methods spanning classical coresets, gradient-based selectors, distribution matching, GBDT-native sampling, and gain-path variants. The broad 17-method audit uses a lighter diagnostic grid, while a fuller five-budget, five-seed core grid supports the headline performance comparisons for the main methods. The study is deliberately not a new condensation algorithm or a state-of-the-art claim. It is a benchmark and diagnostic analysis: when does a landscape score predict downstream behaviour, when does it fail, and what does that imply for using condensed datasets in GBDT workflows?

Contributions.

- **The SLD diagnostic.** We introduce the Split-Landscape Discrepancy, a no-candidate-retraining metric that quantifies how faithfully a condensed dataset reproduces the GBDT split-gain landscape of the full data under a fixed prediction state (Section III).
- **Systematic comparative analysis.** We present a 17-method diagnostic audit for GBDT-on-tabular settings and a separate five-method core grid over 26 datasets, five budgets, and five seeds, measuring both downstream predictive performance and landscape fidelity (Sections IV and V).
- **Three empirically grounded insights.** A single landscape score is method-level rank-correlated with AUROC at Spearman $\rho = -0.87$ in the small-to-mid-size diagnostic grid, making it useful for screening and family-level ordering rather than top-1 selection. Within-dataset trends are generally negative after controlling for dataset scale. Coverage and predictive fidelity are decoupled: a $(1 - 1/e)$-optimal coverage selector is among the weakest downstream performers, explained by a reference-chain decomposition into structure and leaf-estimate gaps. Structure is robust to perturbation below a margin-linked threshold; the dose is a fraction of the split margin, not a fraction of intentionally corrupted splits (Section V).
- **Scope boundaries.** We document where the lens does not extend: regression tasks (structure-dominated failure), hyperparameter-rank transfer (near-zero Jaccard), and no-candidate-retraining selection of the single best method (top-3 rate 0.44), measured under the same protocol (Section VI).
- **Reproducibility repository.** The benchmark protocol, SLD implementation, processed tables, seeds, command lines, and plotting scripts are released at https://github.com/VIVAN-RANGRA/sld-gbdt-condensation.

II. Related Work

a) Dataset distillation and condensation.: Dataset distillation [1] and its successors optimise a small synthetic set so that a model retrained on it matches one trained on the full data, using gradient matching [2], distribution matching [10], or trajectory matching [11]. Comprehensive surveys [12]–[14] emphasize downstream evaluation, mostly on image benchmarks, and do not provide a GBDT-specific split-landscape fidelity metric. Extending distillation to tabular data has attracted recent attention [5], [15]–[17], but these works retain the retrain-and-evaluate paradigm. Our contribution is orthogonal: rather than a new condensation method, we introduce a no-candidate-retraining diagnostic that measures whether *any* condensed set, regardless of the method that produced it, preserves the GBDT split-gain landscape.

b) Coreset selection.: Classical coreset methods select real training examples by geometric or gradient-based criteria: herding [3] matches distribution moments; k-center [18] minimises the maximum coverage distance; the CRAIG coreset method [4] and Gradient Matching (GradMatch) [19] match gradient norms via submodular maximisation; Error L2 Norm (EL2N) and Gradient Normed (GraNd) [20] use early-training gradient norms. All of these methods were designed for and evaluated on differentiable models. Translating their selection signals to the GBDT setting is non-trivial because GBDTs are not differentiable with respect to the training set; we include adapted variants in our 17-method benchmark and show that their landscape fidelity, measured by SLD, predicts their tabular AUROC. The GBDT-native subsampling methods Gradient-based One-Side Sampling (GOSS) [9] and Minimal Variance Sampling (MVS) [21] operate per boosting round rather than producing a static condensed set; MVS is the theoretically closest prior work, as it explicitly minimises the variance of the split-scoring estimator. Data valuation [22], [23] and influence-function methods offer related but complementary perspectives on data contribution; they often require retraining on many subsets or differentiable-model assumptions and are not directly applicable in their standard form to histogram GBDTs.

c) Why GBDTs for tabular data.: Gradient-boosted decision trees [7]–[9], [24] remain one of the strongest practical model classes for tabular data [25], [26]. This motivates a condensation evaluation framework specific to tree-structured inductive bias. The split-gain criterion [27], [28] aggregates per-bin gradient statistics, and the resulting landscape fully determines which feature and threshold a node selects, a structure that has no direct analogue in differentiable models. Our SLD exploits this structure to build a measurement instrument that is GBDT-mechanism-specific: it explains *why* a condensed set preserves tree behaviour, not merely whether it does.

d) The C^2TC contrast.: The most closely related recent work is C^2TC (Class-Adaptive Clustering for Tabular Condensation) [6] (ICDE 2026), a tabular condensation method that optimises class allocation and feature representation through class-adaptive clustering. Our work is different in intent: C^2TC is a condenser, while SLD is a diagnostic that audits whether any surrogate preserves the GBDT split-gain landscape under a fixed prediction state. We did not include C^2TC in the 17-method comparative grid because its official implementation was not directly compatible with our frozen weighted-surrogate benchmarking interface at the time of grid construction. To avoid leaving this interaction untested, we ran a compatibility audit using the official C^2TC code on the Adult split at budget 200. The generated surrogate achieved AUROC 0.769, but had raw $\text{SLD}_\infty = 1228.21$, relative

$\mathrm{SLD}_\infty = 4.72$, and root agreement $= 0.0$. We interpret this as a compatibility and scope check, not a comparative result: budget 200 and the synthetic C^{2}TC surrogate fall in exactly the regime where our own scale and margin analyses warn that root-level SLD can lose fine AUROC resolution. The run confirms that SLD can audit C^{2}TC-generated surrogates, but it is not a full C^{2}TC baseline.

III. Background and the SLD Diagnostic

A. GBDT Split-Gain Landscape

Let the full training dataset be $D = \{(x_i, y_i)\}_{i=1}^N$ with features $x_i \in \mathbb{R}^F$ and labels y_i. A gradient-boosted decision tree (GBDT) fits an additive ensemble of trees by performing, at each node, a greedy split that maximises the regularised gain [7], [8]. Given first- and second-order loss statistics $g_i = \partial_{\hat{y}}\ell(\hat{y}_i, y_i)$ and $h_i = \partial^2_{\hat{y}}\ell(\hat{y}_i, y_i)$ for each instance i, the gain of partitioning the instance set I at a node into left and right children I_L, I_R is

$$\mathcal{G}(I_L, I_R) = \frac{1}{2}\left[\frac{\left(\sum_{i\in I_L} g_i\right)^2}{\sum_{i\in I_L} h_i + \lambda} + \frac{\left(\sum_{i\in I_R} g_i\right)^2}{\sum_{i\in I_R} h_i + \lambda} - \frac{\left(\sum_{i\in I} g_i\right)^2}{\sum_{i\in I} h_i + \lambda}\right] - \gamma, \tag{1}$$

where $\lambda \geq 0$ is the ℓ_2 leaf-weight regulariser and $\gamma \geq 0$ is the minimum gain threshold [8]. Modern histogram-based GBDTs [9], [24] discretise each feature f into B bins; per-bin sums of (g, h) suffice to evaluate every candidate split.

Lemma 1 (Histogram sufficiency; standard, e.g. [9], [21])**.** *Given the current gradients and Hessians, a histogram GBDT with B bins per feature has its full per-node split-gain landscape determined entirely by the $F \times B$ grid of per-bin (g, h) aggregates. No other information from the training set is required to evaluate* (1) *at every candidate (feature, threshold) pair.*

We define the *split-gain landscape* $\Gamma_D \in \mathbb{R}^{F\times(B-1)}$ as the gain $\mathcal{G}$ that each (feature, bin-boundary threshold) candidate would achieve on dataset D under a fixed prediction state. Lemma 1 is a standard consequence of the histogram GBDT algorithm; we state it here to ground the diagnostic. Its importance is histogram sufficiency: once the prediction state fixes g_i, h_i, Γ_D can be computed in one histogram pass.

B. Split-Landscape Discrepancy

Let $D' = \{(x'_j, y'_j, w'_j)\}_{j=1}^m$ be a weighted selected or synthetic surrogate of D, with $m \ll N$. Synthetic points are binned using the same feature preprocessing and bin boundaries as D; their labels may be hard labels or soft labels converted to the objective's expected gradients. We quantify the structural gap between D and D' via their landscape distance.

Definition 1 (Split-Landscape Discrepancy)**.** *Let the raw landscape error be $\delta_p(D, D') = \|\Gamma_D - \Gamma_{D'}\|_p$. The relative Split-Landscape Discrepancy of order p between dataset D and surrogate D' is*

$$\mathrm{SLD}_p(D, D') \;=\; \frac{\delta_p(D, D')}{\max(\|\Gamma_D\|_p, \epsilon)}, \tag{2}$$

using the same bins, objective, regularisation, base score, and feature preprocessing for D and D'. We use $p = \infty$ (worst-case relative split distortion) and the rank-based split-regret *variant. Root-SLD compares the root landscape. Reference-tree early-node SLD averages over early nodes of a fixed full-data reference tree, using the reference node memberships. Full-path SLD extends the same comparison along deeper reference-tree paths.*

Both variants are computed from histogram statistics under a fixed prediction state, and crucially *no model is retrained on D'*. Figure 1 illustrates how SLD fits into the condensation evaluation pipeline as a no-candidate-retraining alternative to the retrain-and-evaluate loop.

C. Diagnostic Sanity Checks

The diagnostic rests on two standard facts about histogram GBDTs and set coverage. We use them only to motivate what SLD measures; the contribution of the paper is the empirical measurement study across 17 condensation methods.

a) Margin preservation.: Let Δ_v be the gain margin between the best-ranked and second-best-ranked split at node v on D. If

$$\delta_\infty(D, D') < \Delta_v/2, \tag{3}$$

then the argmax split selected by D' coincides with the argmax selected by D at node v, assuming a unique argmax and deterministic tie-breaking. In the dose-response cells satisfying this raw-error margin condition, root-level split agreement equals 1.0 (Section V-C).

b) Coverage guarantee.: The split-coverage objective is the unweighted fraction of full-data high-gain split thresholds, the top-k full-data split candidates used by the coverage selector, whose adjacent bins are represented by at least one selected or weighted surrogate point. It is binary threshold coverage, not gain-weighted path fidelity. As a standard set-coverage objective, it is monotone submodular under a cardinality constraint, so greedy maximisation has the usual $(1 - 1/e)$ guarantee [29]. This defines HistDistill-Greedy: it covers the split landscape near-optimally yet performs poorly by downstream predictive performance (Section V-B). The gap between coverage and predictive fidelity is the central empirical finding of Section V-B.

D. Computing SLD: Algorithm

Algorithm 1 summarises the SLD computation. The key property is that, after the fixed prediction state is available, scoring a candidate requires only one histogram pass and one arithmetic comparison of landscapes; no candidate GBDT is trained. After shared binning, histogram

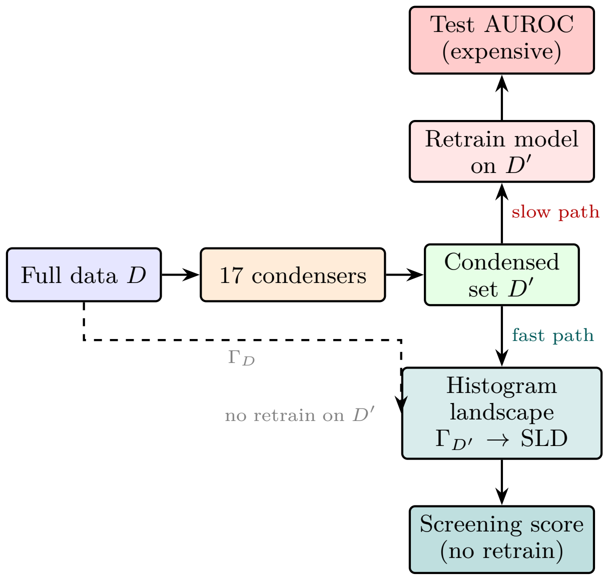

Fig. 1: **SLD evaluation pipeline.** After selecting a condensed set D' from full data D via any of 17 condensers, the fast path computes the candidate's histogram split-gain landscape $\Gamma_{D'}$ under the same fixed prediction state as Γ_D. It provides a cheap screening signal before retraining on each candidate, in contrast to the slow path of retrain-and-test evaluation.

accumulation costs $O(NF)$ for full data and $O(mF)$ for the surrogate; prefix-sum gain evaluation and memory are $O(FB)$ per node.

Algorithm 1 Compute Split-Landscape Discrepancy without candidate retraining

Require: Full dataset D, weighted surrogate D', bins, norm order p, fixed prediction state π

Ensure: $\text{SLD}_p(D, D')$ and split-regret score

1: Compute first- and second-order loss statistics for D under π $\triangleright$ root base score or reference-booster predictions

2: Build histogram $H_D \in \mathbb{R}^{F\times B\times 2}$: for each feature f and bin b, store $\left(\sum_{i:x_{if}\in b} g_i,\ \sum_{i:x_{if}\in b} h_i\right)$

3: Compute landscape $\Gamma_D \in \mathbb{R}^{F\times(B-1)}$ via (1) applied to every candidate split

4: Repeat Steps 1–3 for D' to obtain $\Gamma_{D'}$ $\triangleright$ selected rows reuse weights; synthetic rows are evaluated under π

5: $\text{SLD}_p \leftarrow \|\Gamma_D - \Gamma_{D'}\|_p / \max(\|\Gamma_D\|_p, \epsilon)$

6: **Split-regret:** let $(f^*, b^*) = \arg\max_{f,b}[\Gamma_D]_{f,b}$ and $(\hat{f}, \hat{b}) = \arg\max_{f,b}[\Gamma_{D'}]_{f,b}$;

7: split-regret $\leftarrow \max\left(0, \frac{[\Gamma_D]_{f^*,b^*} - [\Gamma_D]_{\hat{f},\hat{b}}}{\max(|[\Gamma_D]_{f^*,b^*}|, \epsilon)}\right)$

8: **return** SLD_p, split-regret

IV. Experimental Setup

A. Methods Evaluated

We evaluate 17 condensation methods spanning five groups; Table I gives the full taxonomy. **Geometry/classical coresets** include Random sampling, Herding [3], and K-center [18]. **GBDT-native subsampling** methods include GOSS [9] and MVS [21], adapted to produce static condensed sets. **Gradient- and score-based selection** methods include GraNd, EL2N [20], CRAIG-style [4], GradMatch-style [19], and gradient sampling (loss-weighted). **Distribution-matching distillation** is represented by DistributionMatching [10] adapted to tabular features. **Gain/landscape-based methods** form our primary family: HistDistill-Greedy (the $(1-1/e)$-optimal coverage selector), HistDistill-Refined and HistDistill-Density (controlled landscape-fidelity diagnostic baselines), gain sketch, gain-path sketch, and the legacy gain-path refined method. The last three are prior gain-path methods; the paper's main claim is the diagnostic and benchmark, not a new state-of-the-art condenser. All adaptations produce static surrogates: GOSS/MVS select the top scored instances under the fixed GBDT state; GraNd, EL2N, CRAIG, and GradMatch use their published scoring or matching rules; DistributionMatching runs on processed tabular features and shares full-data bin edges for SLD scoring. Short names follow the original methods: GOSS is Gradient-based One-Side Sampling, MVS is Minimal Variance Sampling, GraNd is Gradient Normed, EL2N is Error L2 Norm, CRAIG is a gradient-matching coreset method, DM abbreviates DistributionMatching, and HD abbreviates HistDistill.

B. Datasets and Protocol

Our experiments use public tabular datasets drawn from OpenML [30] and the Penn Machine Learning Benchmarks (PMLB) [31]; several entries originate from UCI-style benchmark datasets. The full inventory appears in Table X. The repository lists the exact OpenML/PMLB identifiers and processed split checksums. The core study is on binary classification. For each dataset we condense the training set to five budgets, $\{10, 25, 50, 100, 200\}$ rows, and repeat every (dataset, method, budget) configuration over five random seeds. Across the 26 binary datasets and the five core methods this gives 3,250 retrained-model evaluations for headline performance (Table II) and 650 landscape-fidelity measurements, one per dataset, method, and budget.

The broader 17-method analysis uses the same 26 datasets, budgets 25/50, and three seeds, giving 156 cells per method for Figure 2. The decomposition comparison (Table V) isolates the two reference-chain failure modes; its absolute AUROCs are not compared to the pooled headline values. Additional studies include the 1,872-cell split-noise robustness analysis, multiclass classification, regression, hyperparameter optimisation (HPO) transfer, and a large-scale analysis on Covertype/SUSY/HIGGS with 400-row condensed sets; Sections V-E and VI give details.

TABLE I: The 17 condensation methods. †: adapted from differentiable settings.

Family	Method	Signal / citation
Geometry	Random Herding K-center	Uniform Moments [3] Coverage [18]
GBDT-native	GOSS† MVS†	Grad-norm [9] Min-var [21]
Gradient	GraNd† EL2N† CRAIG† GradMatch† Grad. samp.†	Norm at init [20] Error L2 [20] Submod. [4] Grad match [19] Loss-weighted
Distrib.	DistributionMatching	Distr. match [10]
Gain / landscape	HD-Greedy HD-Refined HD-Density Gain sketch Gain-path sk. Legacy gp	$(1-1/e)$ coverage Landscape fidelity Land.+density Gain sketch Gain-path sketch Gain-path refined

C. Metrics

Downstream performance. Binary classification: AUROC. Multiclass: one-vs-rest macro-AUROC. Regression: R^2. All scores use the model retrained on the condensed set and evaluated on the held-out test set.

Landscape fidelity. The relative SLD_∞ is defined in (2). Figure 3 uses raw (un-normalised) ℓ_∞ error to show scale confounding; raw values are not comparable across datasets. Split-regret normalises the gain cost of following the surrogate's argmax split over the full-data optimum. Let $g^* = [\Gamma_D]_{f^*,b^*}$ be the full-data optimal gain, and $\hat{g} = [\Gamma_D]_{\hat{f},\hat{b}}$ the gain of the surrogate's preferred split $(\hat{f},\hat{b}) = \arg\max[\Gamma_{D'}]$:

$$\mathrm{regret}(D, D') = \max\left(0,\ \frac{g^* - \hat{g}}{\max(|g^*|, \varepsilon)}\right). \tag{4}$$

Error decomposition. Three models form a reference chain: the full-data model with AUROC A_r (structure reference); the condensed-set tree with leaves refit on full data, AUROC A_ℓ; and the condensed-set model with its own leaves, AUROC A_d.

$$\epsilon_{\mathrm{struct}} = A_r - A_\ell, \qquad \epsilon_{\mathrm{leaf}} = A_\ell - A_d. \tag{5}$$

Leaf-population mismatch $\mathrm{pop}(D') = \|p^* - p_{D'}\|_1$ measures how representatively D' fills the model leaves, where p^* and $p_{D'}$ are the empirical leaf-occupancy distributions on D and D'.

D. Learners and Reproducibility

The reference landscape and primary downstream learner use histogram XGBoost. Robustness checks additionally use LightGBM, CatBoost/fallback histogram boosting, random forests, and multilayer perceptrons (MLPs). All datasets are public and preprocessing is deterministic: categorical features follow the repository scripts, numerical features use shared full-data bin edges, and splits use stratified seeds where labels permit. The repository (linked in the contributions) contains seeds, hyperparameters, package versions, processed tables, and plotting code needed to reproduce all reported results.

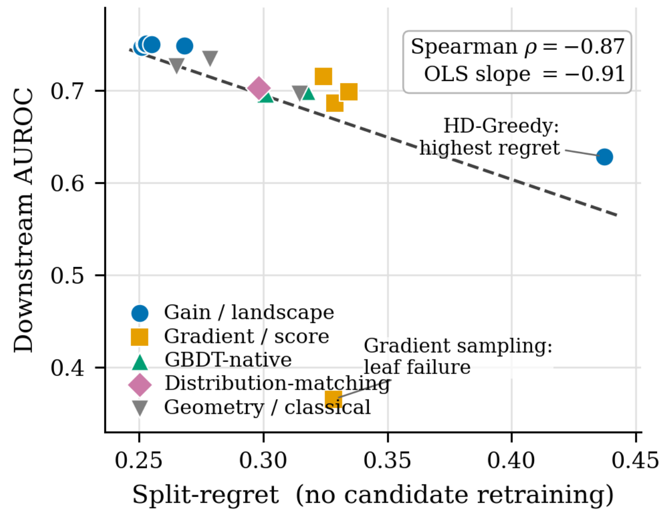

Fig. 2: **No-candidate-retraining method audit.** Each point is one condensation method. Lower split-regret, computed before candidate retraining, tracks higher downstream AUROC over 156 dataset–budget–seed cells (Spearman $\rho = -0.87$, OLS slope -0.91).

TABLE II: Headline performance and diagnostics for five core methods (3,250 cells). AUROC uses mean±95% CI over dataset clusters; avg. rank lower is better. Raw SLD_∞ is diagnostic only and is not a cross-dataset fidelity score; speedup is relative to full-data retraining on the same dataset/learner.

Method	AUROC	Rank	Raw SLD_∞ diag.	Speedup
Full data	0.812 ± 0.062	–	–	1.0×
Legacy gain-path	0.743 ± 0.057	2.60	99.4	3.9×
HistDistill-Refined	0.742 ± 0.057	2.68	93.6	3.5×
HistDistill-Density	0.742 ± 0.057	2.65	93.7	3.5×
Herding	0.731 ± 0.054	3.44	174.1	3.8×
Random	0.724 ± 0.056	3.63	138.3	4.0×

V. Results

A. Does a Training-Free Score Predict Downstream Performance?

A single split-regret score, computed without retraining any candidate, orders condensation methods by their eventual downstream AUROC. The reason is mechanistic. Split-regret measures how much gain the booster forfeits when it follows the surrogate's preferred splits instead of the full-data splits, so a method that misplaces high-gain thresholds is penalised in direct proportion to the structure it destroys. This produces a method-level ordering in the small-to-mid-size diagnostic grid (Figure 2 and Table III).

One method breaks the trend in an informative way. Gradient sampling earns low split-regret yet collapses to 0.366 AUROC, because its failure is in leaf-value estimation rather than split structure (Table V), so we treat it as a diagnosed failure mode rather than noise. The method-level association is not driven only by this point or by

TABLE III: Seventeen-method diagnostic audit. Values are method means ±95% CI over dataset clusters, over 156 dataset–budget–seed cells. Lower split-regret and higher AUROC are better. EL2N and GraNd coincide after rounding under the fixed-state adaptation used here.

Method	AUROC	Regret
Gain-path sketch	0.748 ± 0.062	0.251 ± 0.079
Legacy gain-path	0.751 ± 0.062	0.252 ± 0.079
HD-Density	0.751 ± 0.061	0.253 ± 0.073
HD-Refined	0.750 ± 0.062	0.255 ± 0.077
Random	0.726 ± 0.060	0.265 ± 0.064
Gain sketch	0.749 ± 0.062	0.268 ± 0.081
Herding	0.734 ± 0.059	0.279 ± 0.082
DM	0.703 ± 0.062	0.298 ± 0.075
MVS	0.697 ± 0.058	0.301 ± 0.068
K-center	0.697 ± 0.062	0.315 ± 0.079
GOSS	0.700 ± 0.058	0.317 ± 0.073
GradMatch	0.715 ± 0.057	0.324 ± 0.086
Grad. samp.	0.366 ± 0.067	0.328 ± 0.078
EL2N	0.686 ± 0.056	0.329 ± 0.071
GraNd	0.686 ± 0.056	0.329 ± 0.071
CRAIG	0.699 ± 0.061	0.335 ± 0.082
HD-Greedy	0.629 ± 0.050	0.437 ± 0.106

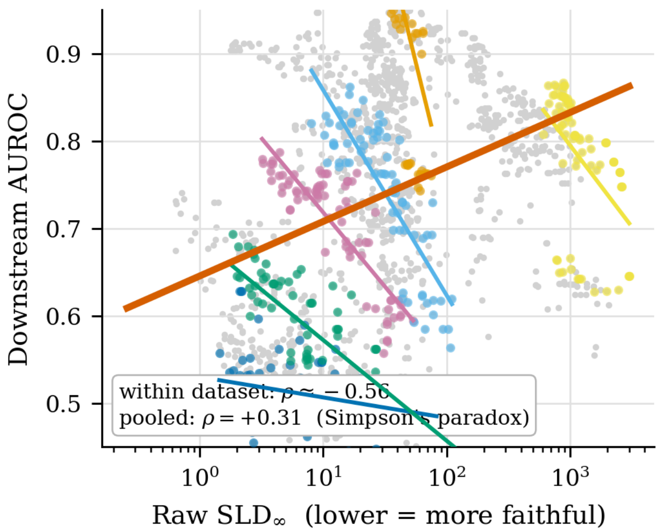

Fig. 3: **Within-dataset SLD trend.** Coloured lines are within-dataset fits on log-scale raw SLD; the average within-dataset trend is negative. The pooled fit reverses because dataset scale confounds raw SLD and AUROC.

HD-Greedy: excluding both leaves Spearman $\rho = -0.85$ over the remaining 15 methods.

Within a single dataset, a more faithful landscape reliably predicts a more accurate retrained model; the relationship inverts only when datasets of different scale are pooled. Holding a dataset fixed and sweeping the budget, lower raw landscape error tracks higher AUROC. Pooling across datasets flips the sign, because larger datasets inflate both the absolute landscape error and the achievable AUROC at once, a textbook Simpson's paradox (Figure 3). The practical consequence is a usage rule rather than a defect: raw SLD is a within-dataset, within-budget comparison signal, and the normalised split-regret, not raw SLD, is the quantity to compare across datasets.

B. Why Does Coverage-Optimal Selection Fail?

A selector that provably maximises split-threshold coverage is among the worst condensers for downstream accu-

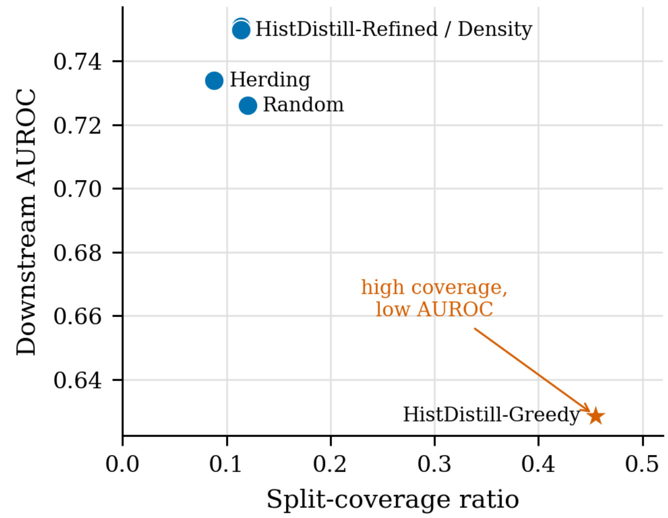

Fig. 4: **Coverage is not predictive fidelity.** HD-Greedy covers the most high-gain split thresholds, but the accurate methods occupy a different region of the coverage–AUROC plane.

TABLE IV: Coverage metrics versus AUROC on the two-budget coverage audit. Cov. is high-gain split coverage; Proxy is coverage divided by the proxy upper bound.

Method	Cov25	Proxy25	Cov50	Proxy50	AUROC
HD-Greedy	0.371	1.000	0.539	1.000	0.629
Random	0.082	0.246	0.159	0.299	0.726
HD-Refined	0.075	0.223	0.152	0.286	0.750
HD-Density	0.075	0.223	0.152	0.286	0.751
Herding	0.059	0.183	0.118	0.229	0.734

racy. HD-Greedy is built to cover as many full-data high-gain thresholds as possible, and it attains the standard $(1 - 1/e)$ guarantee for greedy submodular coverage: it reproduces 37% of high-gain thresholds at budget 25 and 54% at budget 50, against at most 16% for every other method. Broad coverage, however, optimises the wrong objective. Downstream AUROC is driven by the few gain-weighted splits that dominate the prediction, and spreading a tiny budget across many thresholds dilutes exactly those splits. This is why HD-Greedy falls to the bottom of the ranking while the accurate methods occupy a separate region of the coverage–accuracy plane (Figure 4 and Table IV): unweighted threshold coverage and gain-weighted structural fidelity are different objectives.

A reference-chain decomposition shows that the two weakest methods fail for opposite reasons, which is what makes fidelity, not coverage, the right selection target. We split each method's error into a structure component, the accuracy lost because the condensed set induces the wrong tree structure, and a leaf-estimate component, the accuracy lost to mis-estimated leaf values once the structure is held correct. HD-Greedy's loss is almost entirely structural: its structure error is 9.4%, the highest of any method, while its leaf values are healthy (leaf gap 5.4%). Gradient sampling is the mirror image, with a moderate

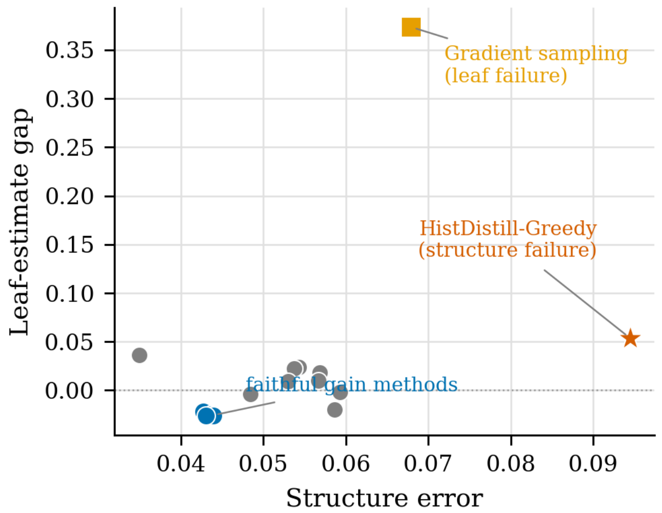

Fig. 5: **Two failure modes.** HD-Greedy is structure-dominated; gradient sampling is leaf-estimate dominated. Faithful gain methods occupy the low-error corner.

TABLE V: Error decomposition on the diagnostic subset used to illustrate both failure modes. Absolute AUROCs differ from the pooled values in Table II and Figure 2. Structure error is reference minus leaf-refit; leaf gap is leaf-refit minus deployed.

Method	**Dep.**	**Leaf**	**Str. Err**	**Leaf Gap**	**Pop.**
HD-Greedy	0.643	0.696	0.094	0.054	0.243
Grad. samp.	0.349	0.723	0.068	0.374	0.133
DM	0.720	0.756	0.035	0.036	0.235
K-center	0.713	0.737	0.054	0.024	0.196
MVS	0.724	0.734	0.057	0.010	0.088
GradMatch	0.734	0.732	0.059	−0.002	0.113
HD-Refined	0.773	0.747	0.044	−0.026	0.060
Legacy gp	0.774	0.748	0.043	−0.026	0.068

Decomposition-subset anchors: full AUROC = 0.826, reference = 0.791.

structure error (6.8%) but by far the largest leaf-estimate gap (37%), consistent with distorted within-leaf statistics rather than poor population coverage (Figure 5 and Table V). A few methods even post slightly negative leaf gaps, meaning their deployed leaves narrowly beat the full-data refit, usually through regularisation or favourable condensed weighting. The selection rule that follows is unambiguous: drive down landscape discrepancy, not coverage.

C. How Robust Is Learned Structure to Landscape Perturbation?

GBDT structure tolerates landscape perturbation up to a margin-linked threshold, after which it degrades gracefully rather than collapsing. The mechanism is the gain-margin condition (3): a split flips only once the injected noise exceeds the gap to its runner-up, and the highest-gain splits, which matter most for accuracy, carry the widest margins and so flip last. Injecting calibrated split-gain noise into the full-data landscape makes the pattern concrete. While the margin condition holds, split agreement and AUROC stay pinned at baseline; past that point agreement falls first and AUROC follows only mildly (Figure 6 and Table VI). This asymmetry explains both halves of the condensation story at once: aggressive compression is safe as long as the surrogate preserves the dominant split margins, and any condenser that corrupts those high-gain splits, as HD-Greedy does, pays a disproportionate accuracy penalty.

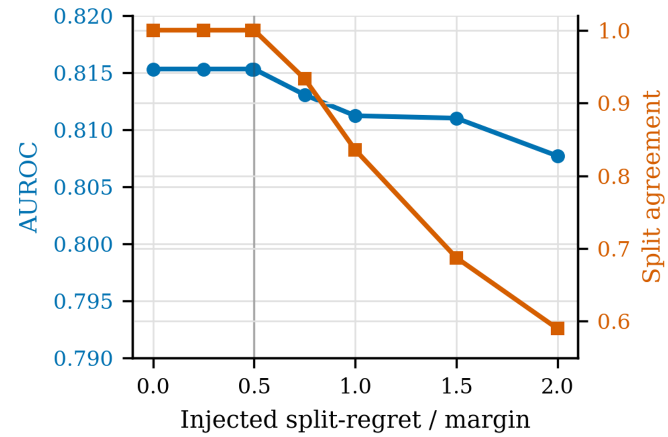

Fig. 6: **Robustness pattern.** Split agreement drops after dose 0.5; AUROC degrades mildly as calibrated split-gain noise exceeds the margin.

TABLE VI: Dose-response: 26 binary datasets, 1,872 cells. Values are mean±95% CI over dataset clusters; eight injected-noise levels are grouped into five displayed rows.

Dose	**AUROC**	**ΔAUROC**	**Agree.**	**Regret**
$\leq$0.50	0.815 ± 0.068	0.000	1.000 ± 0.000	0.000
0.75	0.813 ± 0.069	−0.002	0.933 ± 0.002	0.011
1.00	0.811 ± 0.069	−0.004	0.836 ± 0.004	0.030
1.50	0.811 ± 0.069	−0.004	0.687 ± 0.008	0.069
2.00	0.808 ± 0.070	−0.008	0.590 ± 0.011	0.101

D. What Does the Most Faithful Method Buy?

The most faithful methods match the strongest prior selector at a fraction of the training cost, but they do not beat it. On the controlled core grid, HistDistill-Refined and HistDistill-Density reach the legacy gain-path selector's accuracy while attaining the lowest landscape discrepancy among the core methods (Table II). The match is statistical, not rhetorical: a dataset-clustered paired test places the Refined-versus-Legacy difference within noise of zero, while clearly separating Refined from the weaker Herding and Random baselines (Table IX). The payoff is therefore not higher accuracy but lower cost. Condensed retraining runs several times faster than full-data retraining at these budgets, and the diagnostic itself adds only one histogram pass. We accordingly present the refined family as parity at the top with better fidelity, not as a new state of the art.

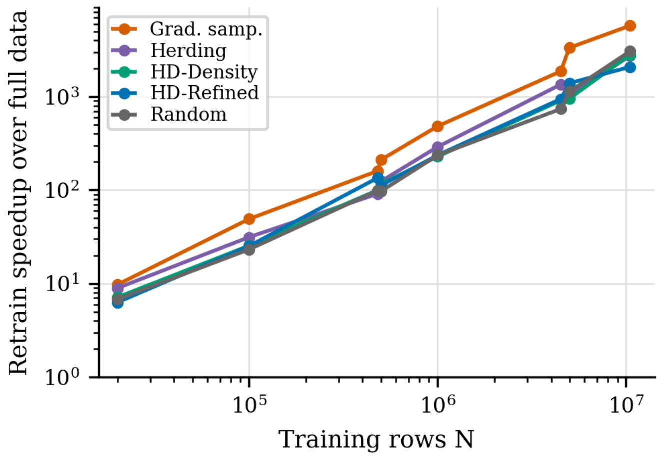

Fig. 7: **Large-scale retraining payoff.** Log-log speedup for 400-row condensed retraining on Covertype, SUSY, and HIGGS. Points are single runs at observed dataset sizes, so no run-to-run CI is claimed for this scale audit.

TABLE VII: Large-scale audit at the largest training size per dataset. ρ is Spearman correlation over the five audited methods; negative ρ means lower split-regret aligns with higher AUROC. Each row is a single large-scale run.

Dataset	Rows	ρ	Top speedup
Covertype	0.48M	−0.36	160×
SUSY	4.50M	−0.97	1,864×
HIGGS	10.50M	0.11	5,734×

E. Does the Diagnostic Hold at Scale?

Condensation's cost advantage grows sharply with dataset size, but at very large N the rankings among reasonable condensers converge and the cheap root-level score loses resolution. A large-scale experiment with fixed 400-row condensed sets shows the retraining speedup growing from modest factors at tens of thousands of rows to several orders of magnitude at the million-row scale (Figure 7 and Table VII). Predictive ordering tells a subtler story. Random holds the best condensed-set point estimate on all three giant datasets, but the three-dataset paired comparison is underpowered, so the honest reading is convergence among reasonable condensers rather than a genuine Random win (Table VIII). This is the regime boundary: the refined family that dominates Random across the small-to-mid-size datasets no longer separates from it once N is enormous.

The mechanism is the same gain margin behind Figures 3 and 6. On the largest dataset, several methods register zero root split-regret because the dominant root split is saturated, so leaf estimates and deeper structure, not the root choice, decide AUROC. Root-level regret still rejects severe failures, but it can no longer finely rank reasonable surrogates once the top margin is cleared. The diagnostic therefore degrades gracefully from a ranker into a filter as N grows.

TABLE VIII: Large-scale AUROC at the largest training size. Full-data AUROCs are 0.897, 0.873, and 0.803 for Covertype, SUSY, and HIGGS. Random wins the condensed point estimates, but the three-dataset paired audit is underpowered ($p = 0.25$ against HD-Refined and HD-Density); no per-row CI is claimed because the large-scale audit uses one full run per dataset/method.

Method	Cov.	SUSY	HIGGS
Full data	0.897	0.873	0.803
Random	**0.792**	**0.839**	**0.687**
HD-Refined	0.747	0.818	0.648
HD-Density	0.781	0.794	0.641
Herding	0.719	0.780	0.450
Grad. samp.	0.377	0.219	0.364

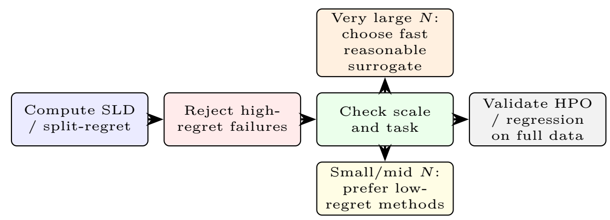

Fig. 8: **Operational workflow implied by the measurements.** SLD is most useful as a cheap screening diagnostic: reject high-regret failures, prefer low-regret methods only when margins are not saturated, and validate HPO, regression, and final deployment choices on full data.

TABLE IX: Dataset-clustered core-method comparisons. Δ is the paired mean AUROC difference over 650 dataset–budget–seed cells; confidence intervals bootstrap the 26 dataset clusters.

Method	Comparator	Δ	95% CI	DS wins
HD-Refined	Legacy gp	−0.00056	[−0.0019, 0.0008]	11/26
HD-Density	Legacy gp	−0.00056	[−0.0016, 0.0005]	9/26
HD-Refined	Herding	0.01183	[0.0037, 0.0204]	18/26
HD-Refined	Random	0.01807	[0.0124, 0.0233]	24/26

VI. Where the Lens Stops

Three settings reveal the boundaries of what split-landscape fidelity can predict. Each was identified by the diagnostic itself, not imposed externally.

Regression: structure mismatch dominates and condensation cannot fix it. Across five regression datasets, deployed condensed models reach R^2 near 0.16–0.17, far below the full-data level of 0.63. The reference-chain decomposition attributes roughly three-quarters of the gap to tree-structure mismatch and only one quarter to leaf-value errors. Refitting condensed-set leaves on full data recovers part of the leaf gap but barely touches the structure gap, and no method closes it at the budgets studied. SLD correctly flags these surrogates as structurally unfaithful; it does not provide a fix. Regression tasks require structure fidelity that small condensed sets cannot deliver (Figure 9).

HPO configuration rankings transfer poorly from condensed to full data. Across 12 binary datasets,

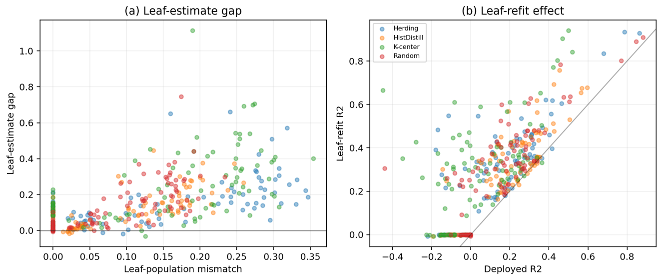

Fig. 9: **Regression decomposition.** Structure loss dominates for every method; leaf refitting helps modestly but does not close the gap to the structure reference.

the top-3 agreement between condensed- and full-data HPO rankings is 4.2% and the top-1 match rate is 8.3%, with a configuration rank correlation near zero (0.15). The median retraining speedup is 1.6×, far too small to compensate. Condensed HPO can find a competitive configuration by chance, but it finds a different one rather than the full-data optimum. Top configurations selected on a condensed set should always be validated on full data before deployment.

Training-free selection of the single best method is unreliable. Split-regret reliably orders method families (Spearman $\rho = -0.87$), but per-dataset selection of the single best method is harder. The top-3 success rate of the SLD-based selector is 43.6% and the top-1 rate is 12.2% across all dataset–budget cells. When the second- and third-ranked methods are close in regret, as is typical, the score is more useful as a rejection filter for clearly poor surrogates than as an oracle.

Across five multiclass datasets the gain-path family improves over Random by 1.3% macro-AUROC, a positive but thin extension of the binary results. Taken together, these findings confirm that SLD is a measurement instrument designed around the GBDT split-gain objective; its signal erodes wherever that objective does not govern the learning.

VII. Conclusion

We introduced the Split-Landscape Discrepancy, a no-candidate-retraining metric that compares full-data and surrogate GBDT split-gain landscapes in one histogram pass, and used it to study 17 condensation methods across 26 tabular datasets. Method-level split-regret correlates with downstream AUROC at Spearman $\rho = -0.87$, showing that the landscape comparison is a reliable screening and family-ranking tool before any candidate is retrained.

The central finding overturns a natural intuition: a selector with a provable $(1 - 1/e)$ coverage guarantee achieves the worst downstream accuracy in the study. Breadth of split-threshold coverage is not a proxy for predictive fidelity, because accuracy depends on preserving the gain-weighted structure of the dominant splits, not on counting reproduced thresholds. The reference-chain decomposition (Equation (5)) operationalises this distinction: coverage failure is a structure failure, not a leaf failure, and each failure mode has a distinct remedy, making the decomposition a practical debugging tool.

The diagnostic reaches its limits at regression tasks, where structure mismatch dominates and small condensed sets cannot overcome it; at HPO transfer, where configuration rankings change even when final AUROC is similar; and at very large scale, where dominant split margins saturate and the score becomes a rejection filter rather than a fine-grained ranker. Within those boundaries, the workflow is simple: compute split-regret to screen out poor candidates, prefer low-regret methods when margins are not saturated, and validate hyperparameter and regression decisions on full data. We release the benchmark and implementation to support future evaluations of new condensation methods.

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Appendix: Dataset Inventory

TABLE X: Expanded 26-dataset binary benchmark inventory. OpenML entries give dataset ID/task ID when a task was used; PMLB entries use exact PMLB identifiers. $*$: focused binary suite. N and F are reported after target removal, benchmark preprocessing, and the 20,000-row cap where applied.

Dataset	ID	N	F	Dataset	ID	N	F
adult*	PMLB:adult	20,000	14	gametes_epi_1000	PMLB:GAMETES_Epistasis_2_Way_1000atts_0.4H_EDM_1_EDM_1_1	1,600	1000
agaricus_lepiota	PMLB:agaricus_lepiota	8,145	22	gametes_epi_20_0.1H	PMLB:GAMETES_Epistasis_2_Way_20atts_0.1H_EDM_1_1	1,600	20
australian*	OpenML d40981	690	14	gametes_epi_20_0.4H	PMLB:GAMETES_Epistasis_2_Way_20atts_0.4H_EDM_1_1	1,600	20
bank_marketing*	OpenML d1461	20,000	16	gametes_epi_3way	PMLB:GAMETES_Epistasis_3_Way_20atts_0.2H_EDM_1_1	1,600	20
breast_w*	OpenML d15/t15	699	9	gametes_het_50	PMLB:GAMETES_Heterogeneity_20atts_1600_Het_0.4_0.2_50_EDM_2_001	1,600	20
credit_approval	OpenML d29/t29	690	15	gametes_het_75	PMLB:GAMETES_Heterogeneity_20atts_1600_Het_0.4_0.2_75_EDM_2_001	1,600	20
credit_g*	OpenML d31/t31	1,000	20	hill_valley_with_noise	PMLB:Hill_Valley_with_noise	1,212	100
diabetes*	OpenML d37/t37	768	8	hill_valley_without_noise	PMLB:Hill_Valley_without_noise	1,212	100
electricity	OpenML d151/t219	20,000	8	kr_vs_kp	OpenML d3/t3	3,196	36
pc1*	OpenML d1068	1,109	21	pc3	OpenML d1050/t3903	1,563	37
pc4	OpenML d1049/t3902	1,458	37	phoneme	OpenML d1489	5,404	5
qsar_biodeg	OpenML d1494	1,055	41	sick	OpenML d38/t3021	3,772	29
spambase*	OpenML d44/t43	4,601	57	tic_tac_toe	OpenML d50/t49	958	9